

Speaker Identification and Voice-Activity Detection in Continuous Meeting Audio

T-03, Phase 1 Report

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1 Narrow Project Question and Scope

The narrowed question is:

In three continuous 300-second excerpts from the AMI Meeting Corpus, how can a simple inference-only pipeline combining pretrained Silero VAD and ECAPA-TDNN detect speech and identify the active speaker among 12 enrolled speakers?

The task uses closed-set speaker identification: every single-speaker interval eligible for speaker-identification accuracy belongs to one of the 12 enrolled identities. This differs from open-set verification, which must accept or reject a claimed identity and handle voices not present during enrollment. The proposed baseline has no unknown-speaker rejection rule, so open-set claims are outside the project scope. Overlap is retained in the continuous recordings as a realistic failure condition, but the baseline is not a full diarisation or speech-separation system.

2 Dataset

2.1 Source and selected subset

The AMI Meeting Corpus is a public multimodal corpus containing approximately 100 hours of recorded meetings, with synchronised audio and several forms of annotation [1, 2]. It is appropriate for T-03 because its natural multi-party recordings include speech, silence, speaker changes, and overlapping speech from multiple speakers. The present project uses the mono **Mix-Headset** WAV recordings for ES2002d, ES2008d, and ES2014d. Audio is downloaded from the official AMI mirror. Speaker timings are obtained as RTTM files from the AMI diarisation setup repository [3].

Each WAV file contains a single-channel waveform sampled at 16 kHz. Each parsed RTTM row contributes the meeting identifier, speech start time, duration, end time, and speaker label. A meeting-qualified label such as ES2002d/FEE005 is used so that every identity is unique across the selected meetings.

Table 1: Selected AMI recordings and annotations.

Meeting	Duration (min)	Audio (MB)	RTTM segments	Speakers
ES2002d	43.74	80.08	774	4
ES2008d	43.76	80.13	757	4
ES2014d	48.52	88.85	695	4
Total	136.02	249.06	2226	12

2.2 Enrollment and evaluation split

For each speaker, the code finds RTTM segments that (i) last from 0.8 to 10 seconds, (ii) have no temporal overlap with another annotated speaker, and (iii) end before 2100 seconds. Candidates are sorted by start time, and the first five are selected. This deterministic rule produces 60 enrollment segments, five per speaker. Their durations range from 0.82 to 9.92 seconds. The latest selected enrollment segment ends at 1826.23 seconds.

The future evaluation material is the interval from 2100 to 2400 seconds in each meeting. Thus, enrollment and evaluation are separated by at least 273.77 seconds. They are temporally disjoint but come from the same meeting sessions; this is a closed-set, session-matched design rather than a cross-session speaker-recognition test.

Table 2: RTTM reference composition of the future evaluation excerpts. Overlap is a subset of speech, not an additional duration.

Meeting	Excerpt (s)	Speech (s)	Silence (s)	Overlap (s)
ES2002d	300	264.79	35.21	50.79
ES2008d	300	238.45	61.55	21.86
ES2014d	300	139.14	160.86	10.24
Total	900	642.38	257.62	82.89

This selection yields 15 minutes of continuous evaluation audio with all four meeting participants present in each excerpt.

3 Reproducible Baseline Setup

3.1 Models and serial pipeline

The pretrained Silero VAD model, version 6.2.1, is used with its default inference settings to return sample-level speech boundaries [7]. The speaker component is `speechbrain/spkrec-ecapa-voxceleb`, an ECAPA-TDNN model pretrained on VoxCeleb data and distributed through SpeechBrain [4, 5, 6]. ECAPA-TDNN maps a variable-length waveform to a fixed-dimensional representation; this implementation produces a 192-dimensional embedding.

The planned Phase 2 pipeline is:

continuous audio $\longrightarrow$ Silero VAD $\longrightarrow$ detected intervals $\longrightarrow$ ECAPA-TDNN embeddings $\longrightarrow$ speaker predictions.

For an enrolled speaker k , five unit-normalised embeddings are averaged and the result is normalised again to form centroid $\mathbf{c}_k$. A detected interval is embedded as $\mathbf{e}$ and compared with every enrolled centroid using cosine similarity,

$$s(\mathbf{e}, \mathbf{c}_k) = \frac{\mathbf{e}^\top \mathbf{c}_k}{\|\mathbf{e}\|_2 \|\mathbf{c}_k\|_2}. \quad (1)$$

Because both vectors are unit-normalised, the implementation uses their dot product. The centroid with the highest score is the top-1 prediction. All 12 centroids are considered, not only the four speakers from the current meeting.

As an exploratory overlap heuristic, Phase 2 will also inspect the margin between the two highest similarity scores. A sufficiently small margin will return both speakers as possible candidates; otherwise only the highest-scoring speaker will be returned. The threshold value is deliberately not fixed in Phase 1. It will be selected and documented before final evaluation and will not be changed in response to test metrics. This heuristic is not presented as a reliable overlap detector because ECAPA-TDNN produces one embedding and was not designed as a multi-speaker classifier.

3.2 Environment and execution

The notebook runs in Google Colab on CPU. Its installation cell is:

```
%pip install -q speechbrain==1.1.0 silero-vad==6.2.1 soundfile==0.14.0
```

The tested environment used PyTorch 2.11.0+cpu and TorchAudio 2.11.0+cpu. The notebook downloads all three audio and RTTM files, parses annotations, selects enrollment segments deterministically, loads the pretrained models, and constructs the centroids. Only requested audio intervals are read from disk, avoiding the need to load all recordings into memory. Since the baseline performs deterministic inference and uses no random sampling, no experiment seed is required in Phase 1.

3.3 Phase 1 checks

The two components were checked independently to confirm that the baseline runs before Phase 2:

- A clean 3.11-second segment not used for enrollment had reference speaker **ES2002d/FEE005**. The top-1 prediction was correct, with cosine similarity 0.7257; the next-highest score was 0.2303. This single example is not reported as accuracy.
- Silero VAD processed a separate 30-second excerpt from seconds 2000–2030 of **ES2002d**, outside the evaluation interval. It returned nine speech intervals totalling 26.35 seconds. No VAD metric or parameter tuning was performed.

4 Phase 2 Evaluation Plan

4.1 Metric 1: Frame-Level VAD F1

This metric evaluates only the timing of speech activity. Each 300-second evaluation excerpt will be divided into non-overlapping 10-ms frames.

A reference frame will be labelled as speech if it overlaps any annotated speech interval and as non-speech otherwise. Similarly, a predicted frame will be labelled as speech if it overlaps any interval detected by Silero VAD.

With speech as the positive class,

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}, \quad (2)$$

$$\text{F1} = \frac{2 \text{Precision Recall}}{\text{Precision} + \text{Recall}}. \quad (3)$$

Here, TP represents frames labelled as speech by both RTTM and Silero VAD, FP represents reference-silence frames incorrectly detected as speech, and FN represents reference-speech frames missed by Silero VAD. This metric measures the agreement between the reference and predicted speech regions; it does not evaluate which speaker is active.

4.2 Metric 2: Top-1 Speaker-Identification Accuracy

Speaker-identification accuracy will be evaluated on the actual speech intervals detected by Silero VAD. For each VAD-detected interval, the evaluation code will find every unique RTTM speaker whose annotated speech overlaps that interval by a non-zero duration.

Each detected interval will then be assigned to one of three groups:

- If the interval does not overlap any RTTM speaker, it is considered a VAD false alarm and is not included in speaker-identification accuracy.
- If the interval overlaps exactly one unique RTTM speaker, it is considered an eligible single-speaker interval.
- If the interval overlaps two or more RTTM speakers, it is treated as an overlapping or mixed-speaker interval and analysed separately.

An interval may be classified as mixed-speaker either because two speakers talk simultaneously or because one VAD interval combines consecutive turns from different speakers.

For every eligible single-speaker interval, the complete waveform produced by the VAD boundaries will be passed to ECAPA-TDNN. Its embedding will be compared with all 12 enrolled speaker centroids, and the centroid with the

highest cosine-similarity score will be the top-1 prediction. The prediction is correct when this speaker matches the only RTTM speaker associated with the interval.

The metric will be calculated as

$$\text{Speaker Accuracy} = \frac{N_{\text{correct}}}{N_{\text{eligible}}}, \quad (4)$$

where N_{correct} is the number of eligible intervals with a correct top-1 speaker prediction, and N_{eligible} is the total number of eligible single-speaker intervals.

Intervals containing no reference speaker are already penalised through the VAD F1 score, while intervals containing multiple speakers are reserved for the separate overlap analysis.

5 Simple Analysis Plan

Phase 2 will inspect the following three categories or slices:

1. **VAD false alarms and missed speech.** Representative false-positive and false-negative regions will be inspected, especially around silence, breaths, hesitations, and low-energy speech.
2. **Short versus long detected intervals.** Eligible single-speaker intervals will be split at their median duration, and top-1 speaker accuracy will be compared between the two groups.
3. **Overlapping or mixed-speaker intervals.** Detected intervals associated with two or more RTTM speakers will be inspected separately. Simultaneous overlap will be distinguished from cases in which one VAD interval merges consecutive speaker turns. For intervals containing exactly two simultaneously active speakers, the analysis will examine whether the similarity-margin heuristic includes both reference speakers among its two candidates and will describe representative successes and failures.

Likely limitations include VAD boundary errors, cross-talk contamination of a single ECAPA embedding, dominance of one voice during overlap, and confusion on short or acoustically similar voices. The analysis will use small summaries and representative examples rather than advanced statistical testing.

References

- [1] J. Carletta, “Announcing the AMI Meeting Corpus,” *ELRA Newsletter*, vol. 11, no. 1, pp. 3–5, 2006.
- [2] AMI Consortium, “AMI Meeting Corpus,” <https://groups.inf.ed.ac.uk/ami/corpus/>.
- [3] BUT Speech@FIT, “AMI diarization setup,” <https://github.com/BUTSpeechFIT/AMI-diarization-setup>.
- [4] B. Desplanques, J. Thienpondt, and K. Demuynck, “ECAPA-TDNN: Emphasized Channel Attention, Propagation and Aggregation in TDNN Based Speaker Verification,” *Proc. Interspeech*, pp. 3830–3834, 2020. <https://doi.org/10.21437/Interspeech.2020-2650>.
- [5] M. Ravanelli et al., “SpeechBrain: A General-Purpose Speech Toolkit,” arXiv:2106.04624, 2021. <https://arxiv.org/abs/2106.04624>.
- [6] SpeechBrain, “Speaker verification with ECAPA-TDNN embeddings on VoxCeleb,” <https://huggingface.co/speechbrain/spkrec-ecapa-voxceleb>.
- [7] Silero Team, “Silero VAD: Pretrained voice activity detector,” <https://github.com/snakers4/silero-vad>.